

Homework Assignment 3A

Power Stage Design

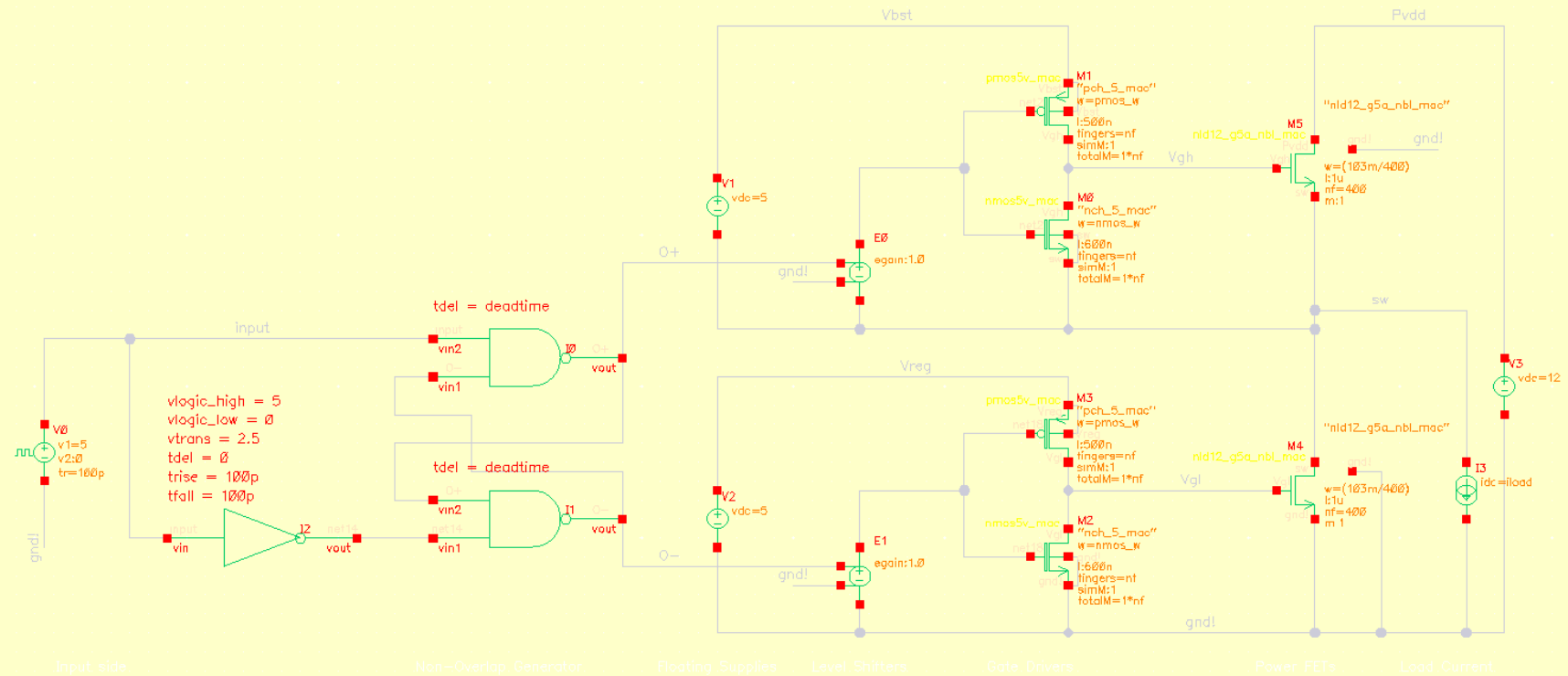
ET4382 - Digital IC Design

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TU Delft

Circuit Overview: NMOS Half-Bridge



Part 1

PowerFET Sizing ($R_{ON} \leq 50 \text{ m}\Omega$ at 150°C)

Part 1: PowerFET Sizing

Goal: Size the nld12_g5a_nbl_mac PowerFET for $R_{ON} \leq 50 \text{ m}\Omega$ at 150C

Method:

DC sweep of total gate width W

$V_{GS} = 5\text{V}$, $V_{DS} = 100\text{mV}$, $T = 150\text{C}$

$R_{ON} = V_{DS} / I_D = 100\text{mV} / I_D$

Result:

At $W = 103\text{mm}$ ($n_f = 400$ fingers): $R_{ON} = 50.02 \text{ m}\Omega$

$I_D = 1.9992\text{A}$ at the target operating point

Key formula:

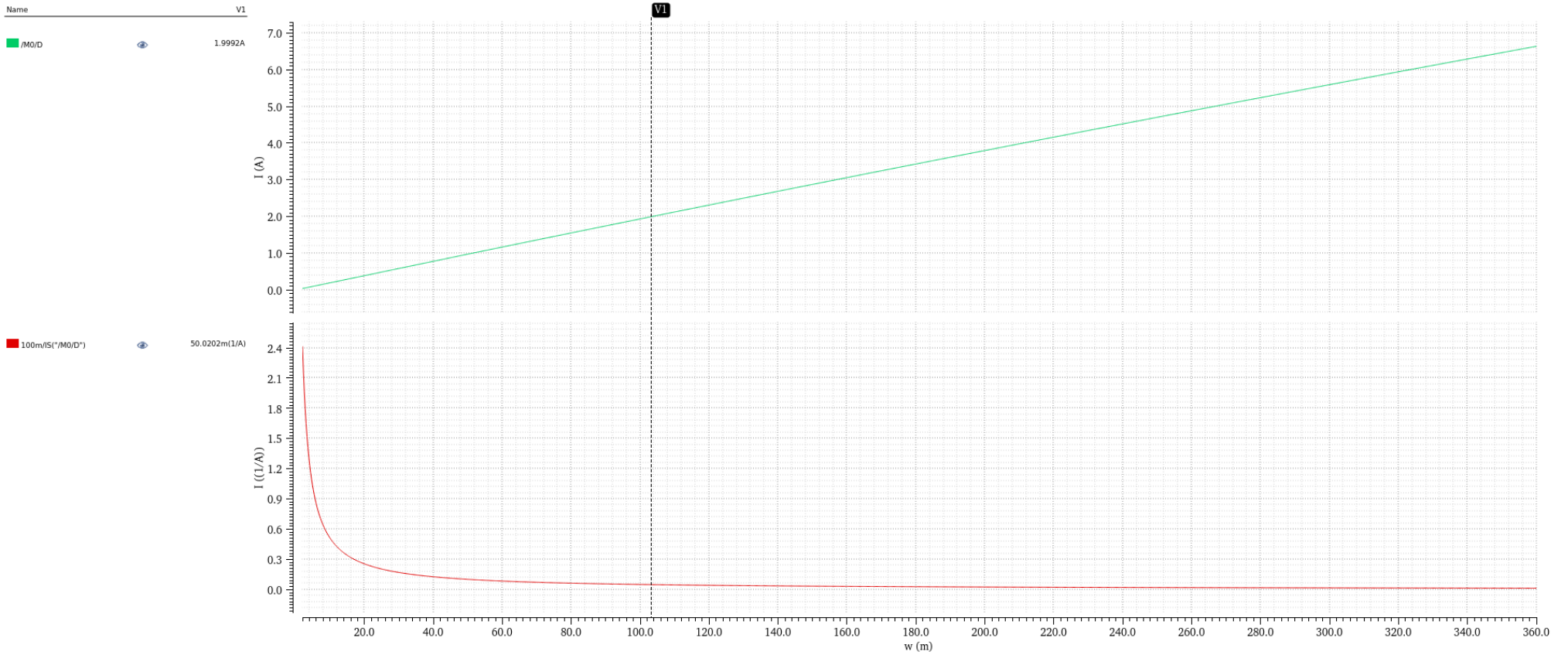
$N_{\text{fingers}} = \text{ceil}(R_{ON,\text{single}} / R_{ON,\text{target}})$

$W_{\text{total}} = N \times W_{\text{finger}}$

Part 1: R_ON vs Width

DC Response

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Top: ID vs W, Bottom: R_ON vs W. Marker at W=103mm, R_ON=50.02 mOhm

Part 2

Threshold Voltage & Gate Charge

Part 2: V_TH Extraction & Gate Charge

V_TH Extraction:

DC sweep: VGS from 0 to 5V, VDS = 100mV, T = 27C

V_TH ~ 0.75V (where drain current starts flowing)

Gate Charge Q_G:

Transient simulation: gate driven 0 to 5V

Gate current integrated over time: $Q_G = \text{integral}(I_G dt)$

$Q_G = 1.247 \text{ nC}$

Gate charge power loss (per half-bridge):

$P_Q = f_{SW} \times 2 \times V_{SUP} \times (Q_{GL} + Q_{GH})$

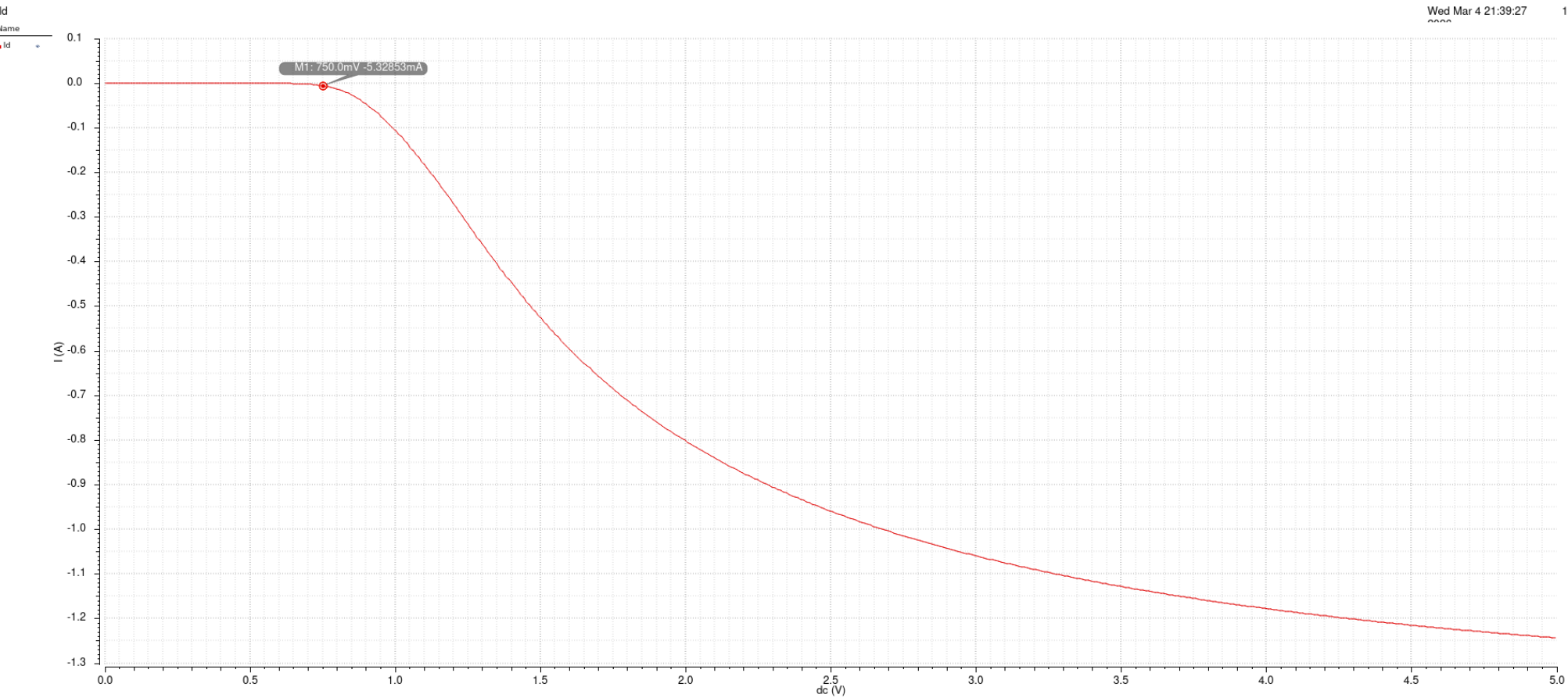
Observations:

Gate current peaks at ~450mA during turn-on

Miller plateau visible in gate voltage ramp

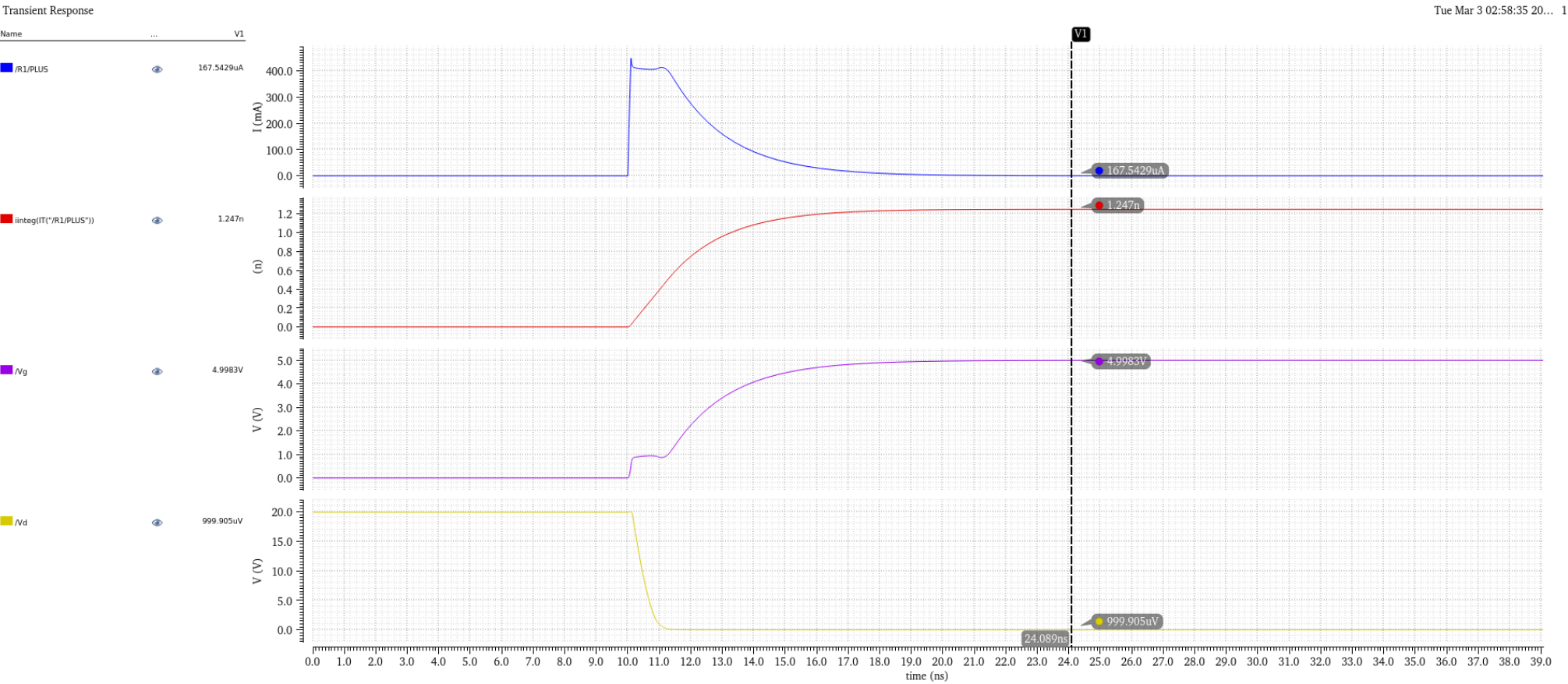
Drain voltage collapses from 20V to ~1mV as FET turns on

Part 2: V_TH Extraction



DC sweep of ID vs VGS. $V_{TH} \sim 0.75$ V at onset of conduction

Part 2: Gate Charge Q_G



Gate current, integrated charge ($Q_G=1.247\text{nC}$), gate voltage, drain voltage

Part 3

Gate Driver Sizing

Part 3: Gate Driver Sizing

Target: ~25ns rise/fall time at PowerFET gate

Driver sizing:

PMOS: pmos5v_mac, W = 3u, L = 500n

NMOS: nmos5v_mac, W = pmos_w x 4 = 12u, L = 600n

Skewed design (W_NMOS >> W_PMOS):

Fast turn-OFF (strong NMOS pull-down) prevents shoot-through

Slower turn-ON (weaker PMOS pull-up) is acceptable with dead time

Measured results:

Rise time (Vgl: 0 to 5V): 19.19ns

Fall time (VgsH: 5V to 0V): 2.61ns

Average gate drive current:

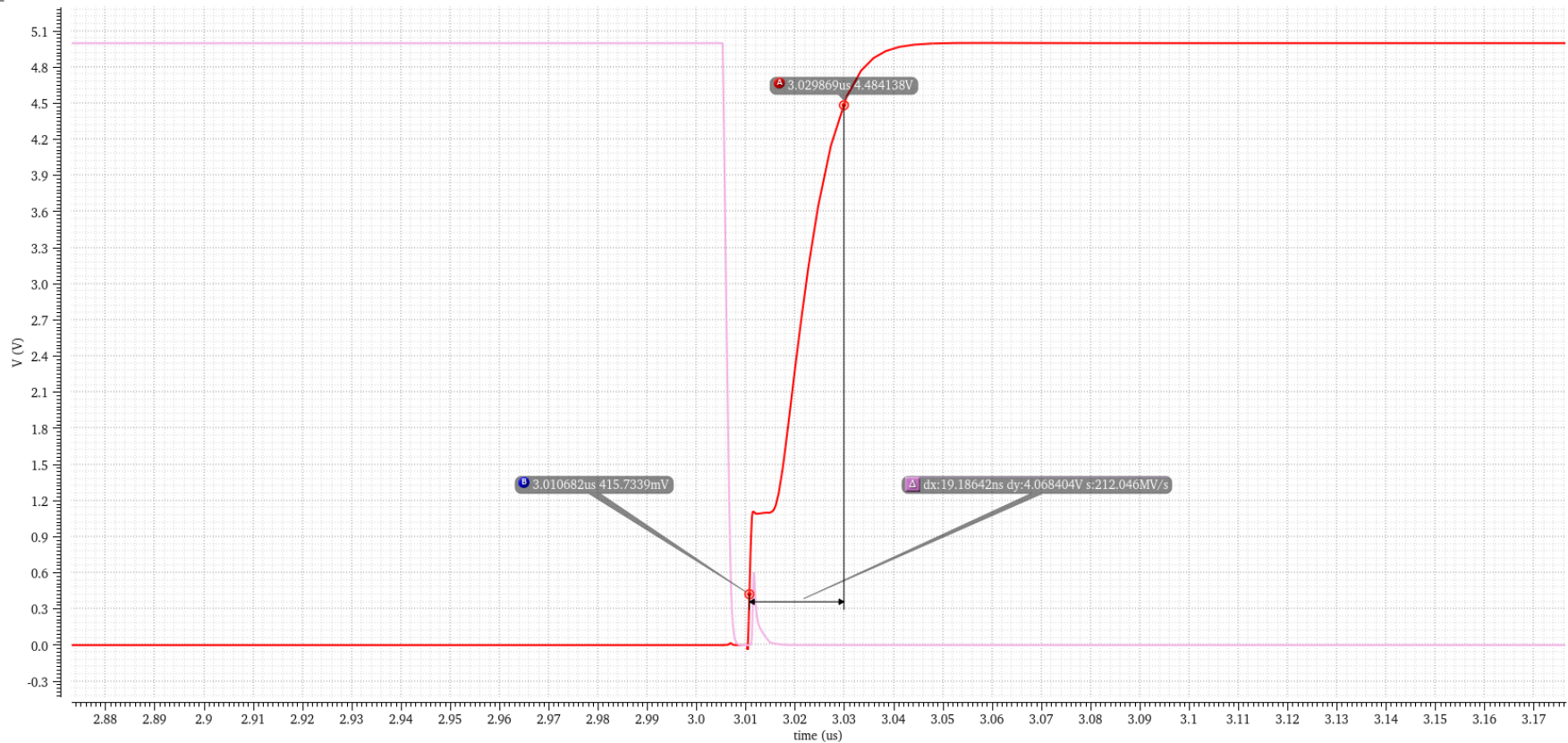
$I_{avg} = Q_G / t_{rise} = 1.247nC / 25ns \sim 50mA$

Part 3: Rise Time

Transient Response

Name	Vis
VT("Vgh") - VT("Vsw")	☒
v /Vgl; tran (V)	☒

1



Vgl rise time = 19.19ns (turn-ON of low-side FET)

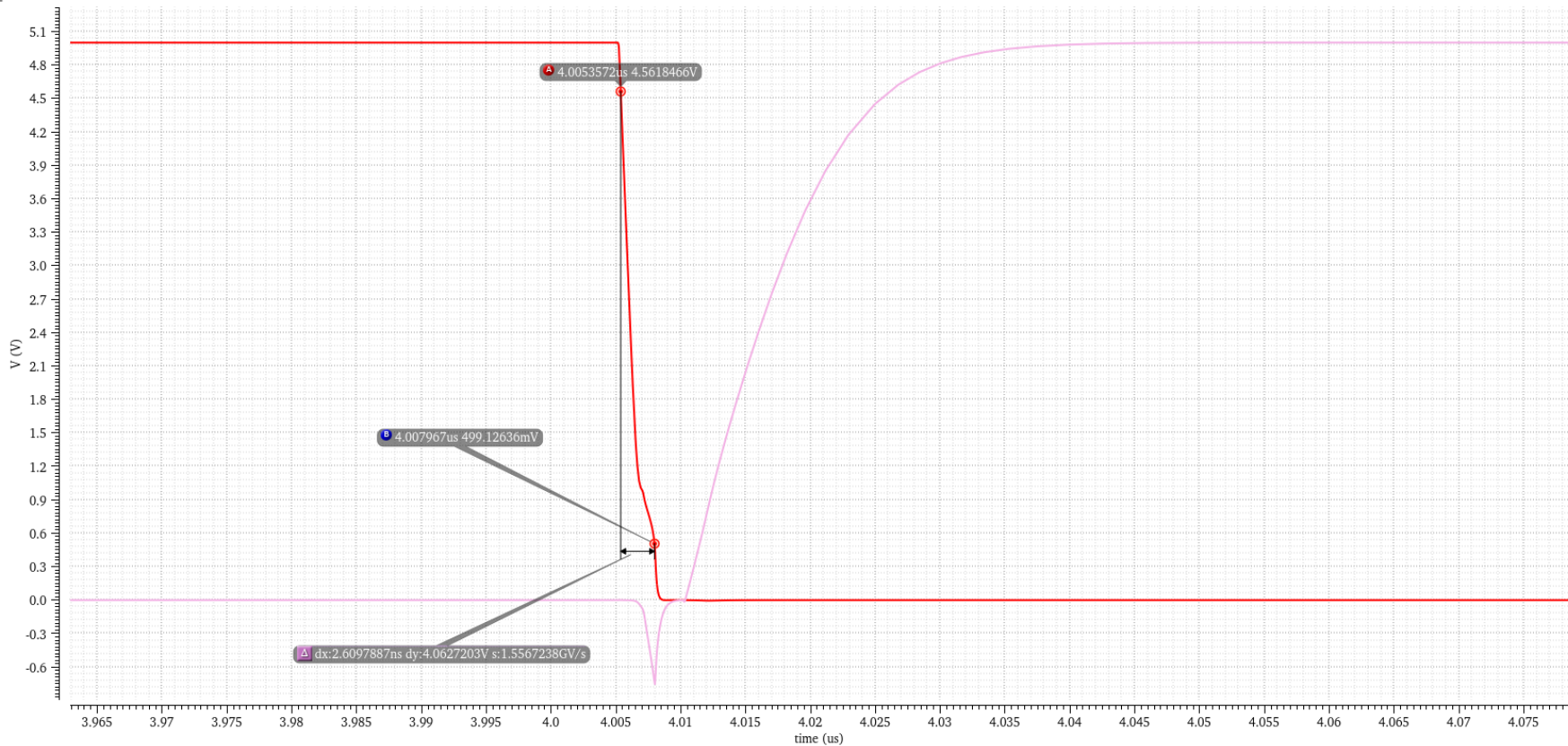
Part 3: Fall Time

Transient Response

Name Vis

VT("Nghi") - VT("sw")

v /Vgi; tran (V)



VgsH fall time = 2.61ns (fast turn-OFF via strong NMOS pull-down)

Part 4

Dead Time Verification

Part 4: Dead Time

Dead time = 5ns (break-before-make)

Implementation:

ahdlLib nand_gate + not_gate with tdel = 5ns

Non-overlap generator ensures both FETs are never ON simultaneously

Measurement:

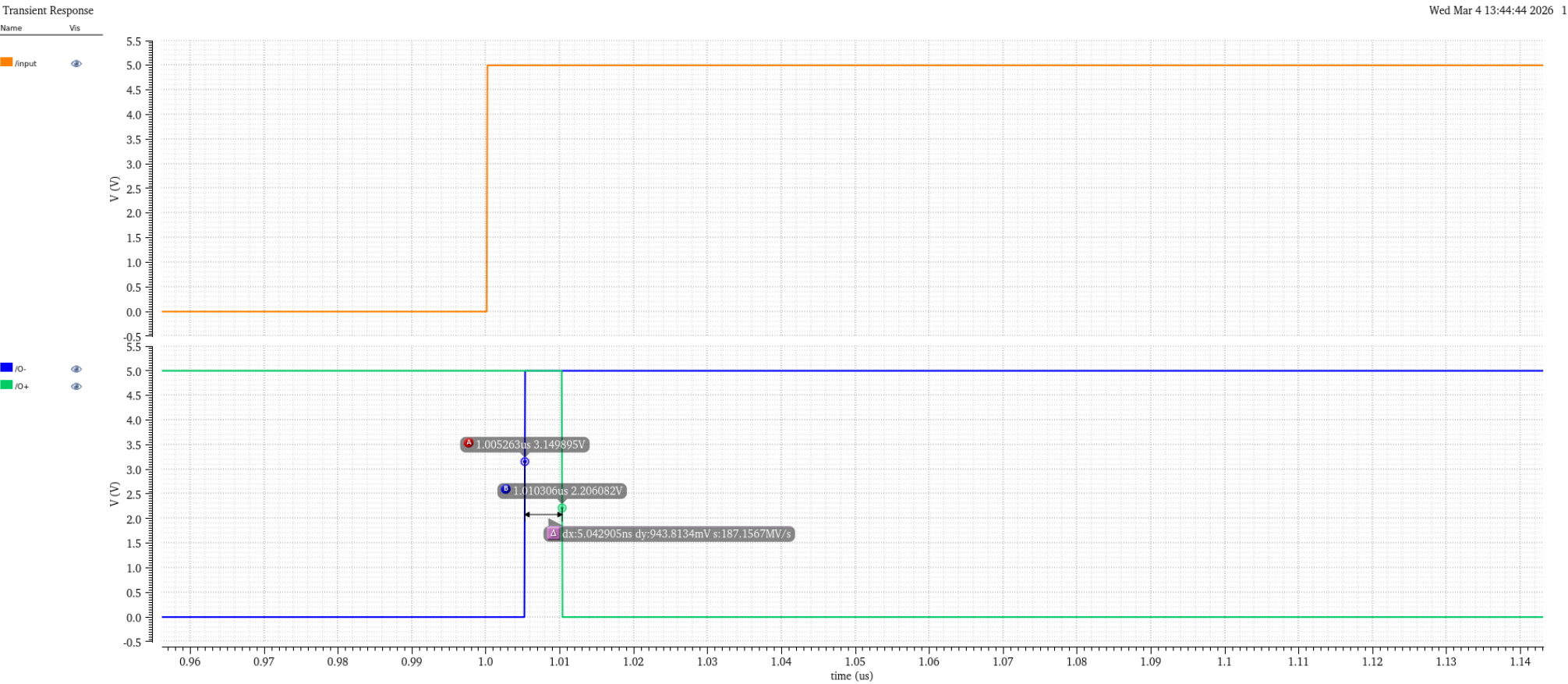
Zoomed to rising edge of /input at ~1.0us

O- falls first, then after 5.04ns, O+ rises

During this gap, both gate drive signals are LOW

Both PowerFETs are OFF (prevents shoot-through)

Part 4: Dead Time Verification



O- falls first, O+ rises 5.04ns later. Both FETs OFF during dead time.

Part 5

Half-Bridge Verification ($\pm 2A$)

Part 5: Soft vs Hard Switching

Soft Switching ($i_{load} = +2A$):

Load current assists the transition

Faster dV_{sw}/dt , body diode conducts after transition

V_{gsH} stays low during dead time (no shoot-through risk)

Hard Switching ($i_{load} = -2A$):

Load current resists the transition

Slower dV_{sw}/dt , body diode conducts during dead time

V_{gsH} rises to $\sim 3V$ due to C_{gd} coupling (shoot-through risk)

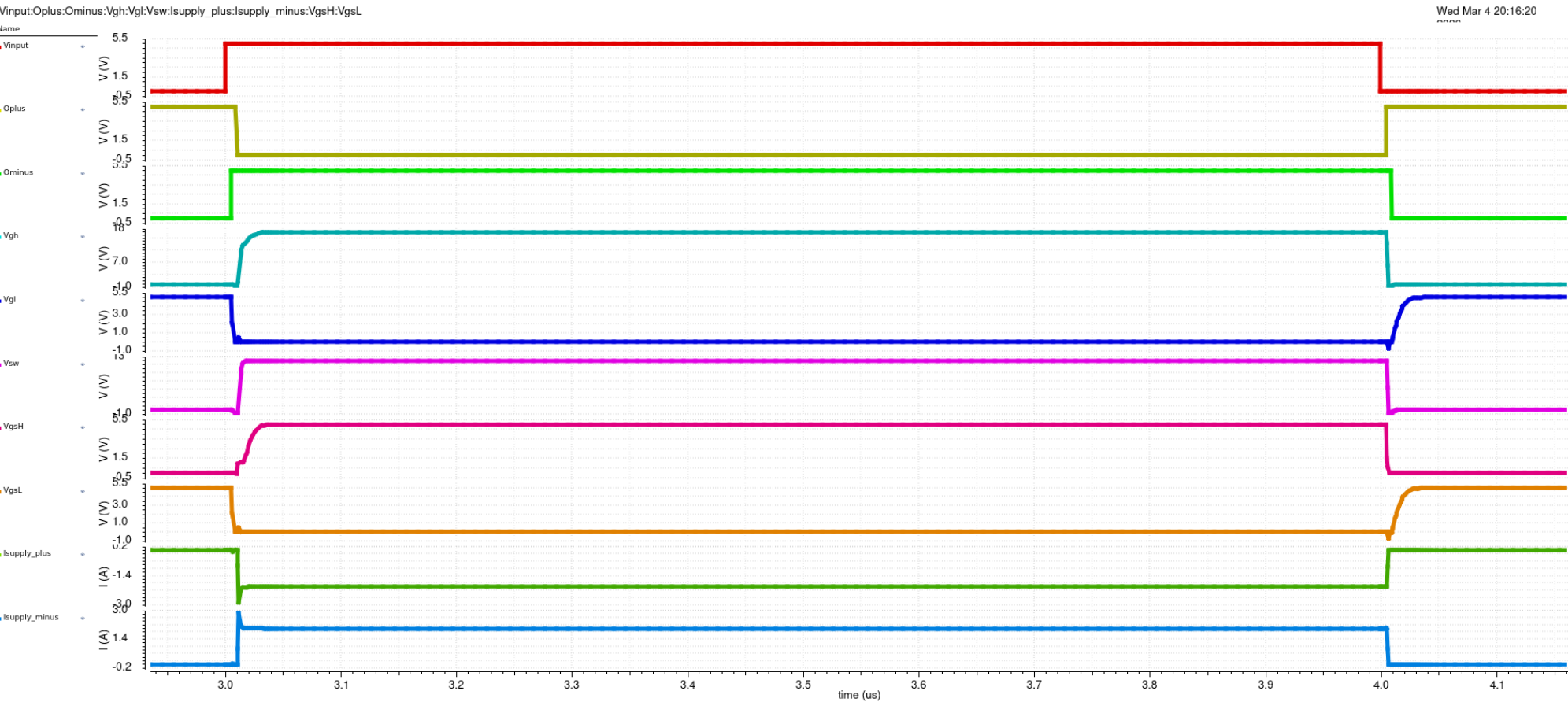
Skewed driver keeps V_{gsH} below V_{TH}

Cgd coupling formula:

$$I_{GD} = C_{GD} \times dV_{sw}/dt$$

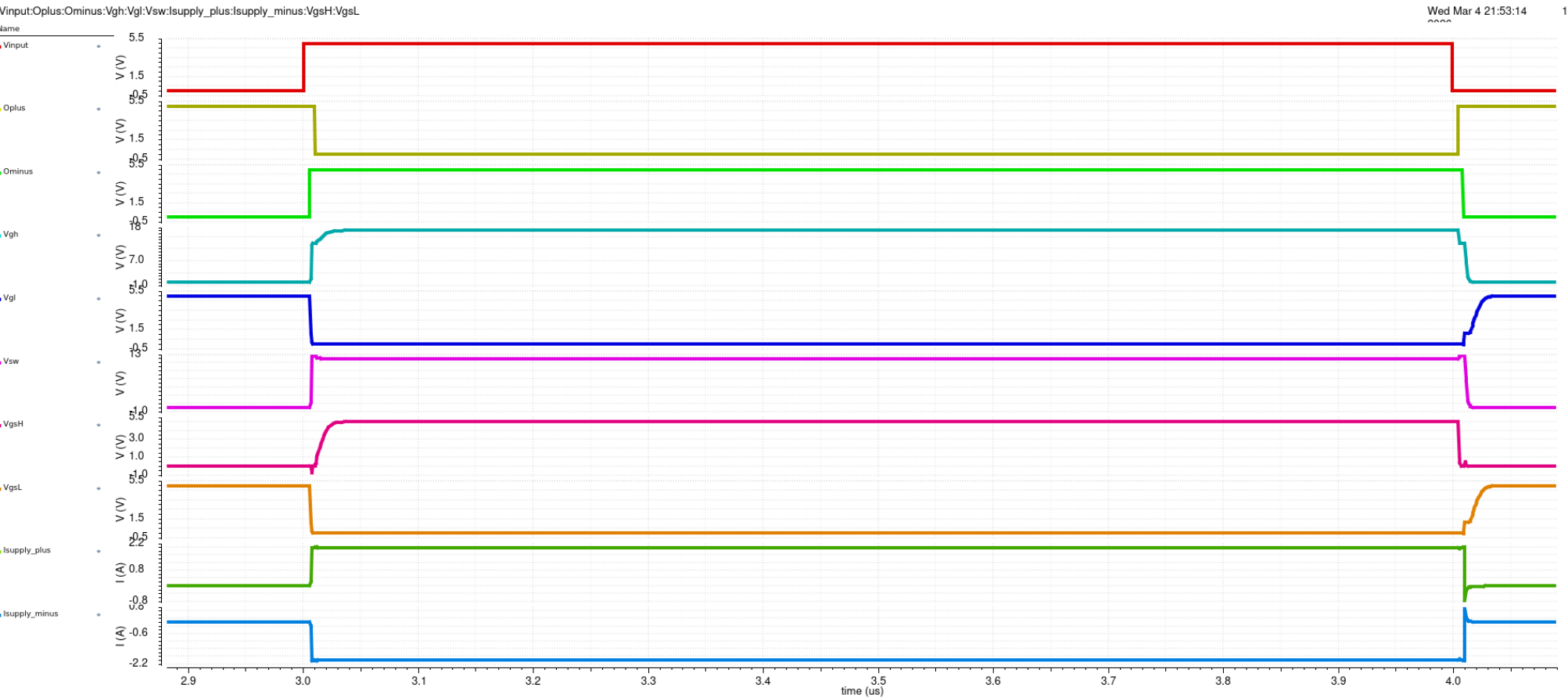
This current pulls V_{GS} of OFF FET above V_{TH} -> shoot-through

Part 5: All Signals (+2A, Soft Switching)



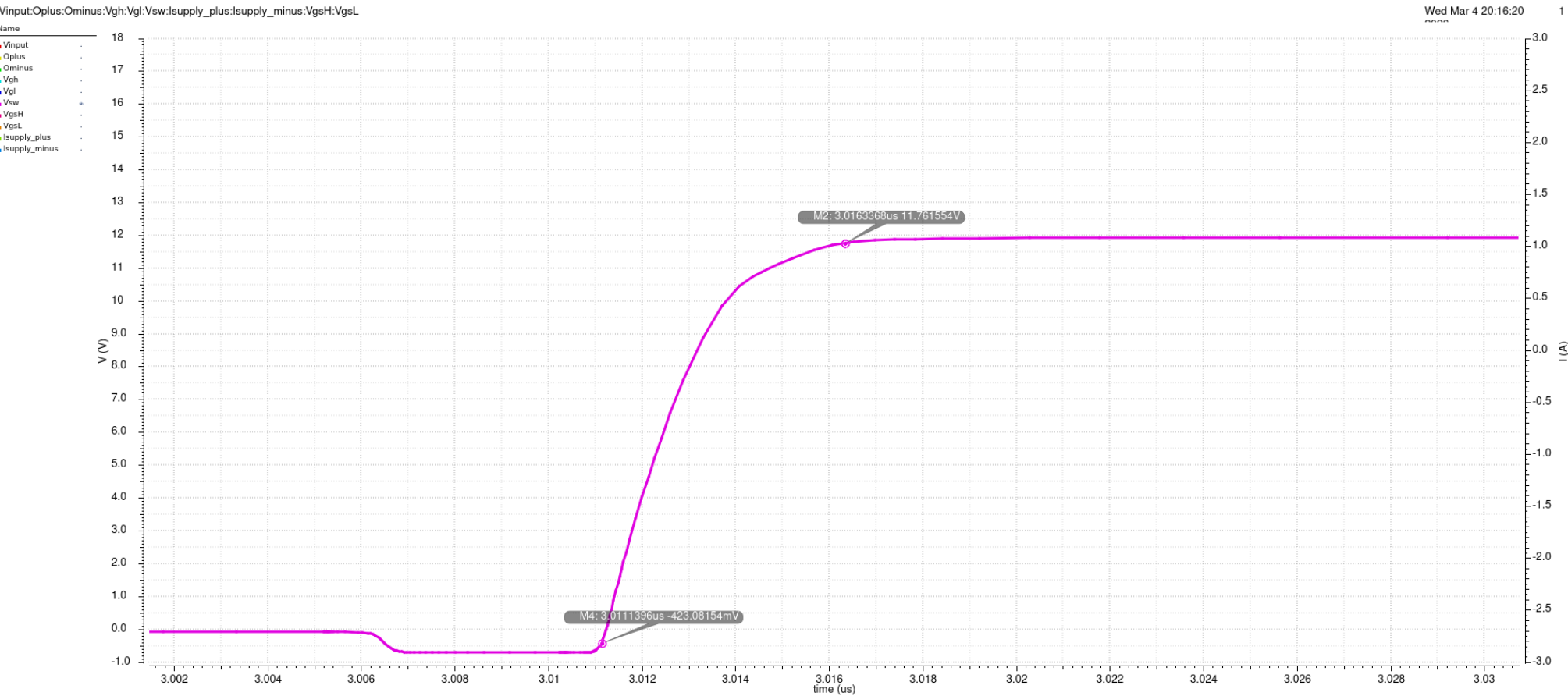
Vinut, O+/O-, Vgh, Vgl, Vsw, VgsH/VgsL, Isupply. Clean switching.

Part 5: All Signals (-2A, Hard Switching)



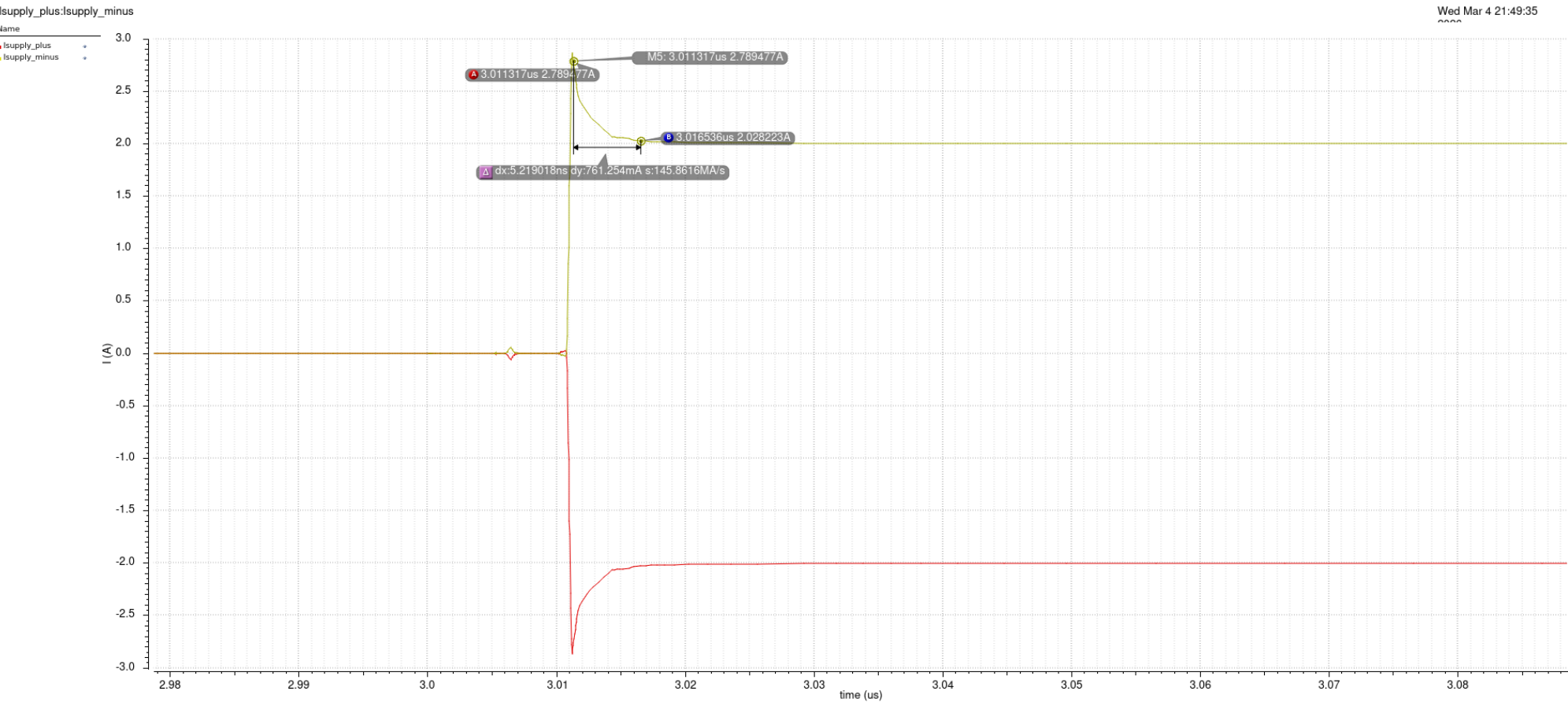
VgsH rises to ~3V during dead time due to Cgd coupling. No shoot-through.

Part 5: Switching Node Zoomed



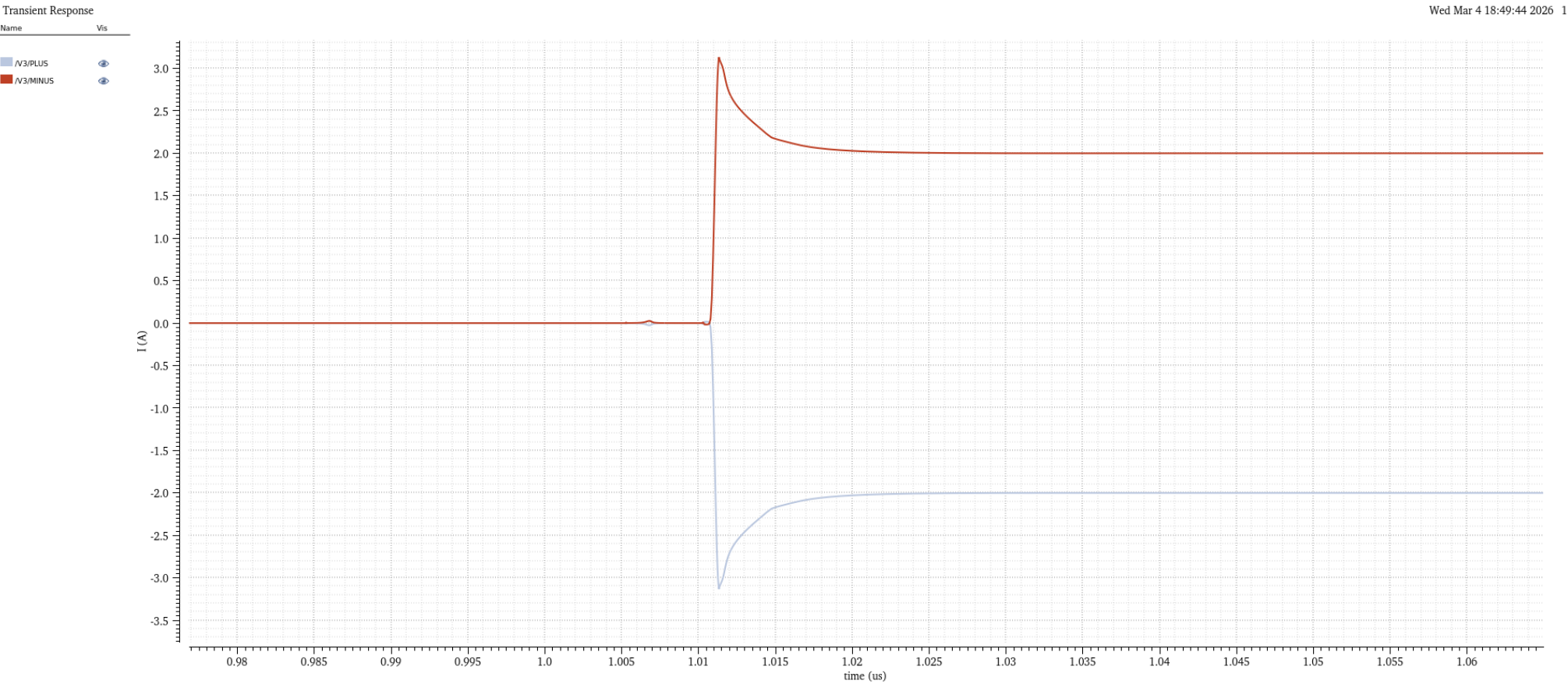
Vsw transition from ~-1V to 12V in ~5ns. Clean, no ringing.

Part 5: Shoot-Through Check



Brief capacitive spike (~2.79A, ~5ns) during transition. No sustained overlap.

Part 5: Load Current at Turn-On



Initial inrush: spikes to ~3.2A, settles to 2A. Normal behavior.

Part 6

BTL Configuration with LC Filter

Part 6: BTL Architecture

Why BTL (Bridge-Tied Load)?

Single half-bridge: sw toggles 0-12V, DC average = 6V offset

BTL: two half-bridges drive load differentially

$V_{diff} = sw_A - sw_B$, swings +/-12V with zero DC offset

AD-PWM (Analog Double-Edge):

Half-bridge A: O+ to high-side, O- to low-side

Half-bridge B: O- to high-side, O+ to low-side (swapped)

When A=HIGH, B=LOW and vice versa

LC Filter:

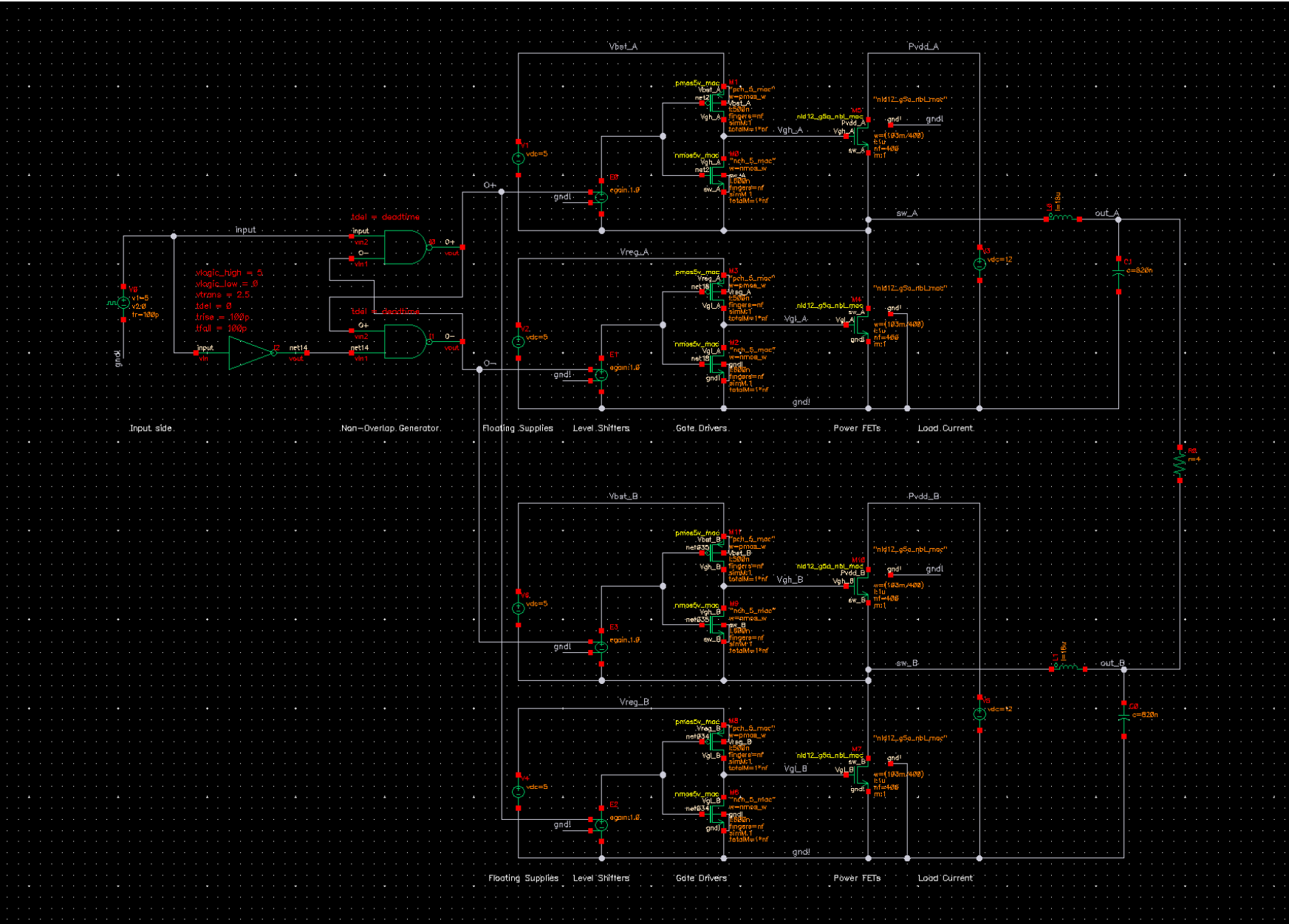
$L = 18\mu H$, $C = 820nF$ per side

$f_0 = 1/(2\pi\sqrt{L*C}) = 41.4 \text{ kHz}$

Passes audio (20Hz-20kHz), blocks 1MHz switching

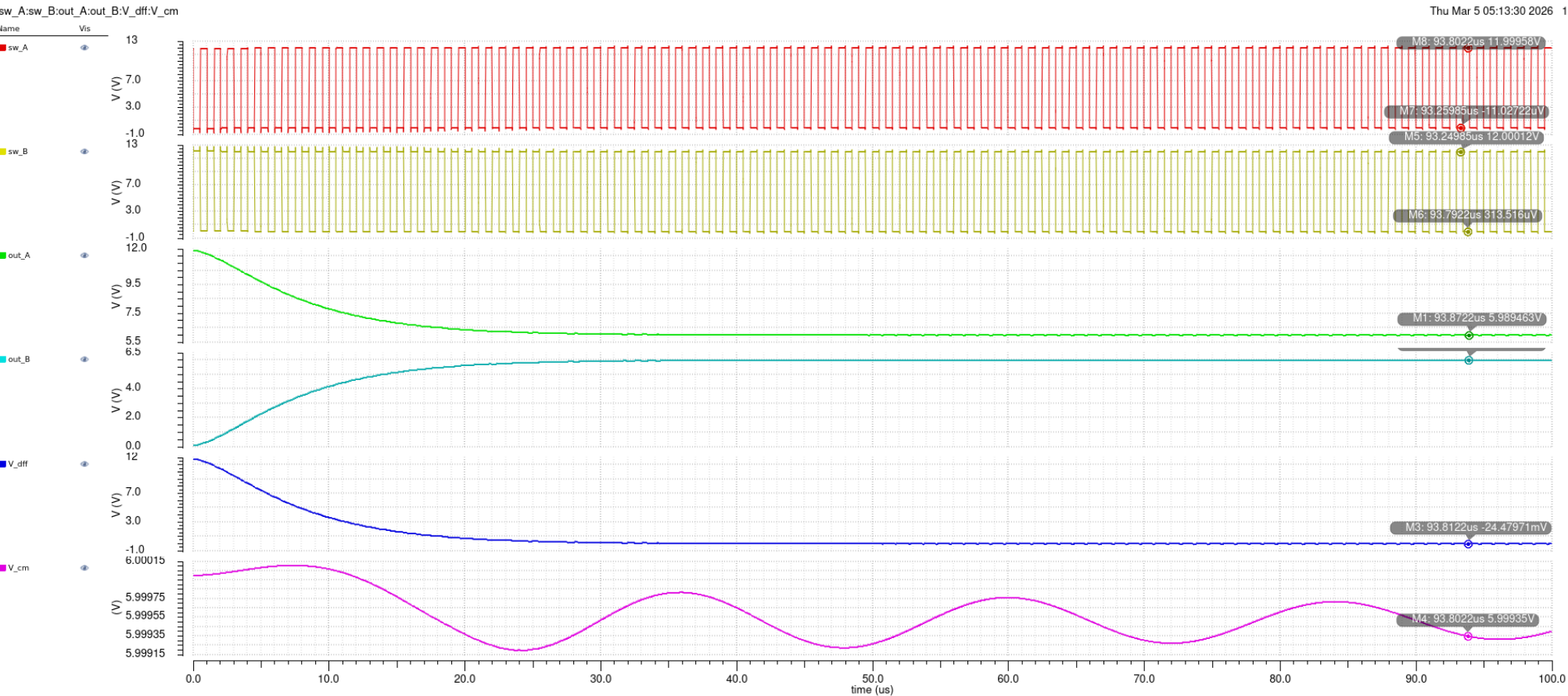
Load: 4 Ohm connected differentially (out_A to out_B)

Part 6: BTL Schematic



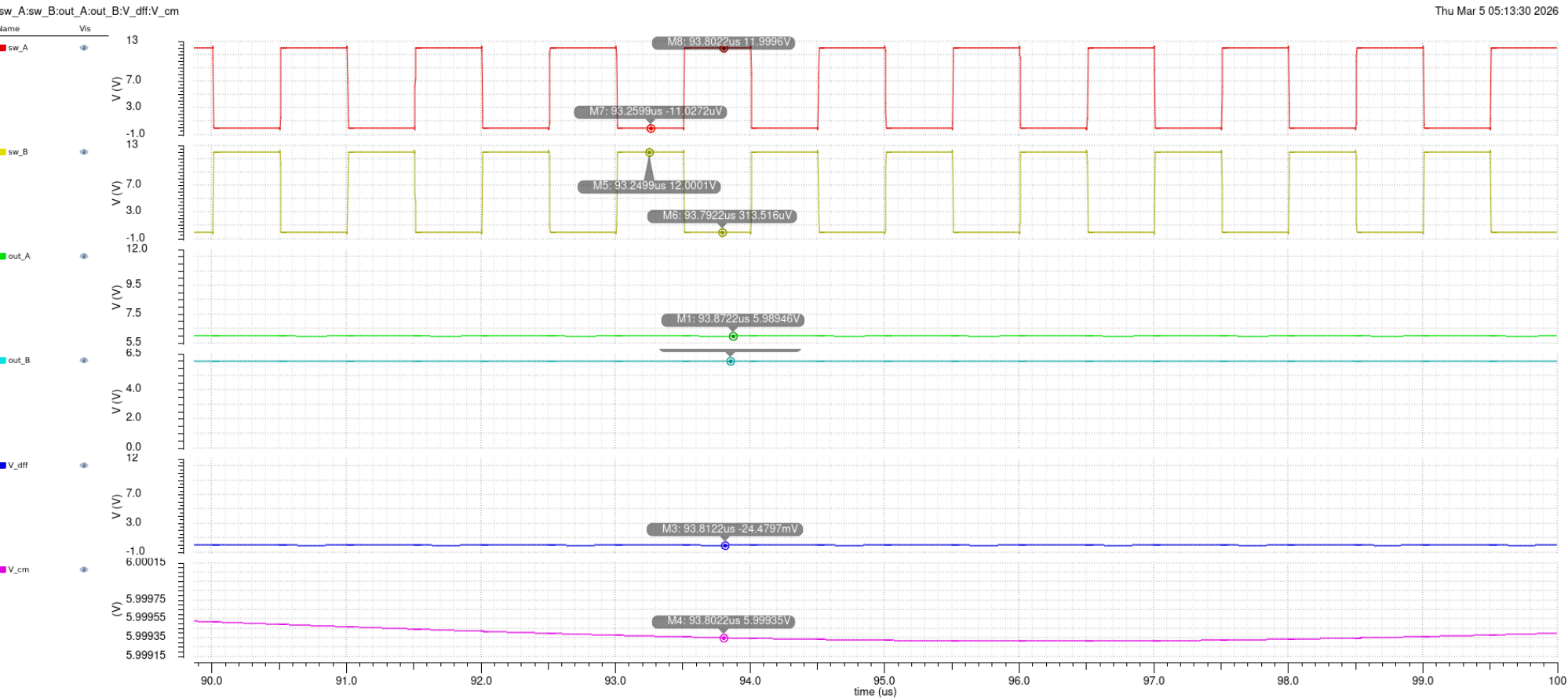
Two half-bridges with swapped O+/O- for AD-PWM, LC filters, differential load

Part 6: BTL at mi=0 (Settling)



Full 100us: sw_A/sw_B inverted, out_A/out_B settle to ~6V, V_diff -> 0V

Part 6: BTL at mi=0 (Settled)



Steady state: $V_{diff} = -24.5\text{mV}$ ($\sim 0\text{V}$), $V_{cm} = 6.0\text{V}$ (constant, AD-PWM property)

Part 6: Efficiency Measurement

Comparator-based PWM for mi sweep:

Triangle carrier (1MHz) + DC signal (mi) + ahdlLib comparator

Duty cycle proportional to modulation index mi

Efficiency formula:

$$\eta = P_{\text{load}} / (P_{\text{sup}} + P_{\text{reg}})$$

$$P_{\text{sup}} = V_{\text{SUP}} \times \text{average}(I_{\text{SUP}}) \quad [12\text{V supply}]$$

$$P_{\text{reg}} = V_{\text{REG}} \times \text{average}(I_{\text{REG}}) \quad [5\text{V gate driver supply}]$$

$$P_{\text{load}} = \text{average}(V_{\text{diff}} \times I_{\text{load}})$$

Parametric sweep: mi = 0 to 0.9 (10 steps)

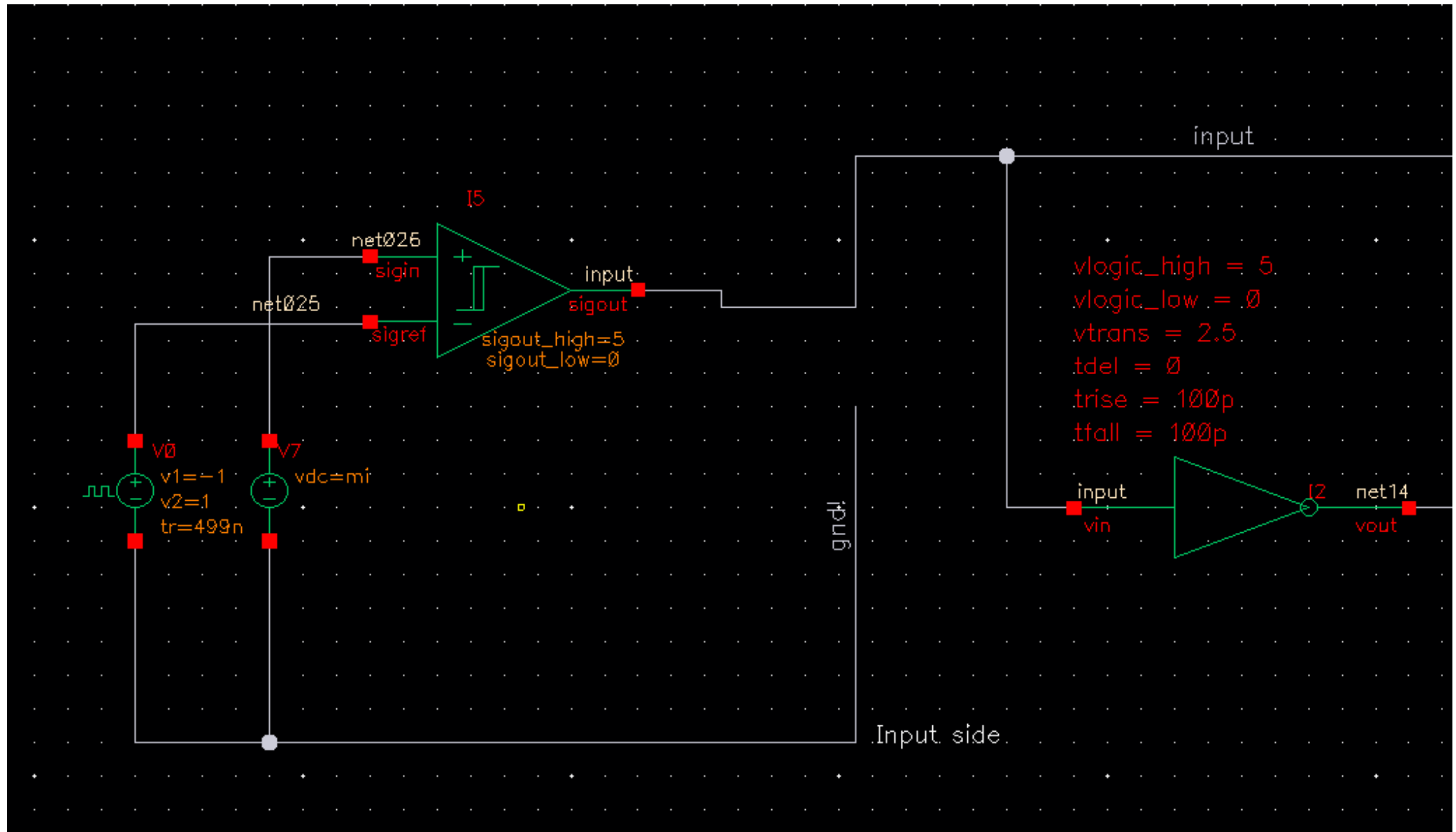
Power loss breakdown:

$$P_{\text{R}} (\text{conduction}) = I^2 \times R_{\text{ON}} \quad \text{-- increases with mi}$$

$$P_{\text{Q}} (\text{gate charge}) = f_{\text{SW}} \times 2 \times V_{\text{SUP}} \times Q_{\text{G}} \quad \text{-- constant}$$

$$P_{\text{X}} (\text{switching}) = f_{\text{SW}} \times t_{\text{x}} \times V_{\text{SUP}} \times I_{\text{OUT}} \quad \text{-- hard switching only}$$

Part 6: PWM Input Stage



Triangle carrier + DC signal (mi) + comparator generates variable-duty PWM

Part 6: Efficiency vs mi (1x Drivers)



P_su, P_reg, P_load, eta vs mi. Efficiency increases with output power.

Part 7

4x Gate Driver Comparison

Part 7: 1x vs 4x Gate Drivers

Tradeoff:

Larger driver -> faster switching -> less P_X (switching loss)

Larger driver -> more driver capacitance -> more P_Q (gate charge loss)

1x Drivers: $p_{mos_w} = 3\mu$, $n_{mos_w} = 12\mu$

4x Drivers: $p_{mos_w} = 12\mu$, $n_{mos_w} = 48\mu$

Expected behavior:

Low m_i : P_Q dominates -> 4x drivers = worse efficiency

High m_i : P_X dominates -> 4x drivers = better efficiency

Crossover point where bigger drivers become beneficial

Simulation result:

1x and 4x efficiency curves are very similar

P_{reg} difference only $\sim 0.3\text{mW}$ (20.2 vs 20.5mW)

Due to ideal components (vcvs level shifters, ahdlLib gates)

Real implementation would show larger differences

Part 7: Efficiency vs mi (4x Drivers)



4x drivers (pmos_w=12u): nearly identical efficiency to 1x drivers

Part 7: 1x vs 4x Comparison

Parameter	1x (pmos_w=3u)	4x (pmos_w=12u)
P_reg (gate driver)	~20.2 mW	~20.5 mW
P_load at mi=0.9	~28 W	~28 W
Switching speed	Baseline	~4x faster
Efficiency curve	Baseline	Nearly identical

Summary

Part 1: PowerFET Sizing

nf=400 fingers, W=103mm, R_ON=50.02 mOhm at 150C

Part 2: $V_{TH} = 0.75V$, $Q_G = 1.247 \text{ nC}$

Part 3: Gate Driver - Rise=19ns, Fall=2.6ns (skewed: $W_N=4 \times W_P$)

Part 4: Dead Time = 5.04ns (break-before-make verified)

Part 5: Half-Bridge verified at +/-2A

Soft switching: clean. Hard switching: VgsH rises to 3V but no shoot-through

Part 6: BTL with LC Filter

$V_{diff} \sim 0V$ at $m_i=0$, $V_{cm}=6V$ constant (AD-PWM property)

Efficiency increases with modulation index (output power)

Part 7: 4x drivers show similar efficiency (ideal component limitation)